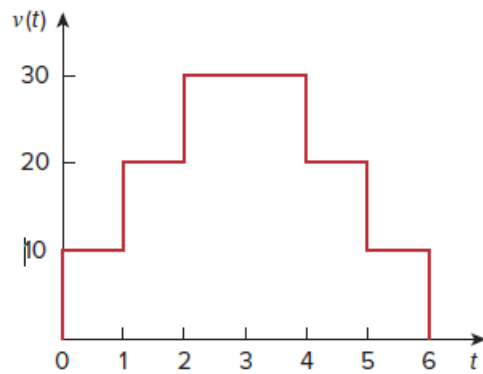


## Problem

One cycle of a periodic voltage waveform is depicted in Fig. 11.66. Find the effective value of the voltage. Note that the cycle starts at  $t = 0$  and ends at  $t = 6$  s.

**Figure 11.66:**



## Step-by-step solution

## Step 1 of 1

$$V_{rms}^2 = \frac{1}{6} \left[ \int_0^1 10^2 dt + \int_1^2 20^2 dt + \int_2^4 30^2 dt + \int_4^5 20^2 dt + \int_5^6 10^2 dt \right]$$

$$V_{rms}^2 = \frac{1}{6} [100 + 400 + 1800 + 400 + 100]$$

$$V_{rms}^2 = 466.67$$

$$\therefore \boxed{V_{rms} = 21.6 \text{ V}}$$